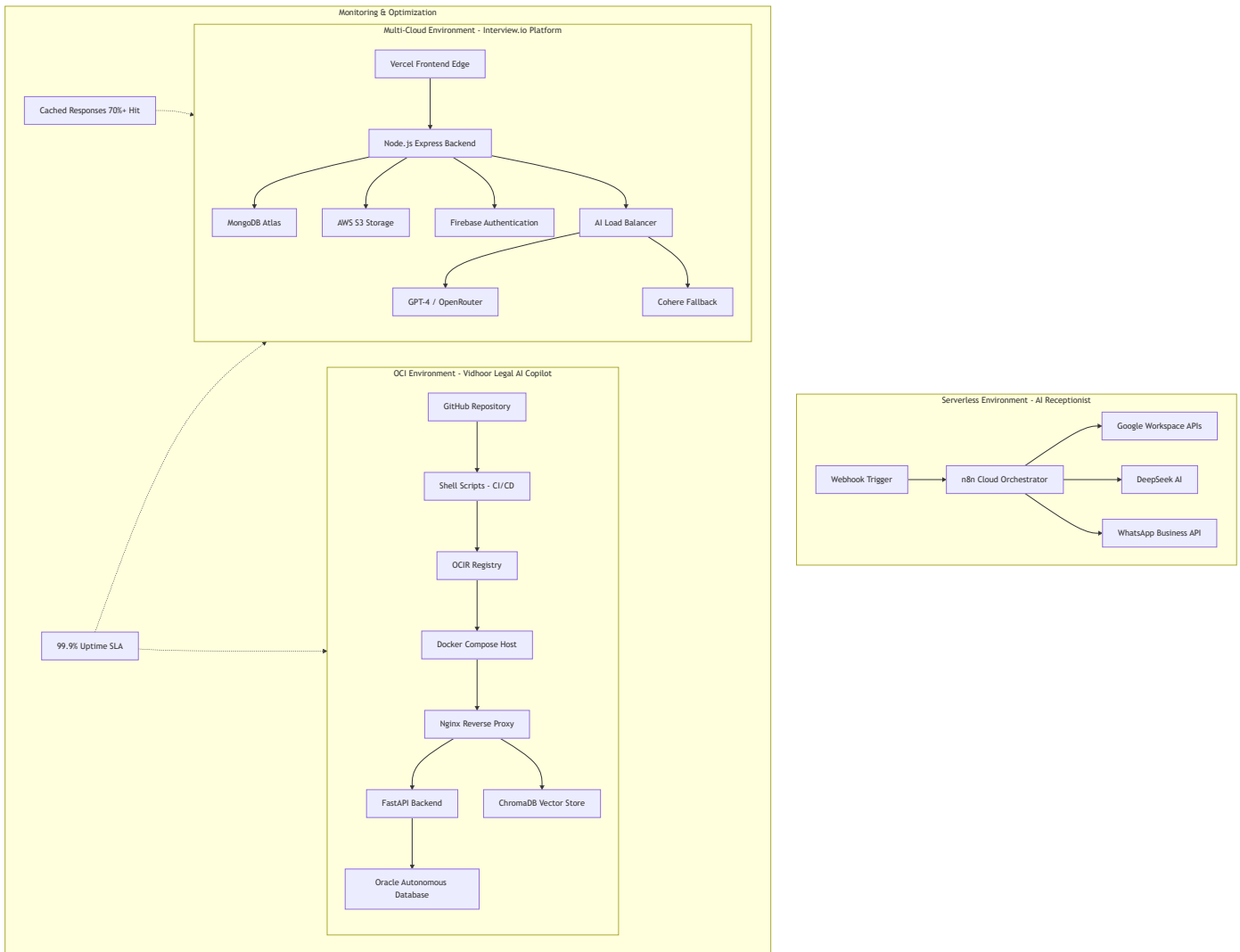


3. Cloud Engineer Portfolio Core

As a Cloud Engineer, I architect the bridge between local innovation and global availability through scalable, secure, and cost-optimized cloud infrastructures. My expertise spans designing multi-container topologies, automating deployments with Infrastructure-as-Code, implementing proactive security protocols, and engineering intelligent cost-reduction strategies that maintain 99.9% uptime while minimizing operational overhead.

Cloud Infrastructure & CI/CD Architecture Matrix



Project Deep-Dives

1. Vidhoor Legal AI Copilot: OCI Cloud & DevOps Automation

Architecting Scalable Infrastructure

- Designed a multi-container distributed backend on **Oracle Cloud Infrastructure (OCI)** with isolated service layers for API, vector storage, and database tiers
- Deployed **FastAPI** backend for RESTful endpoint exposure, integrated with **ChromaDB** for local vector similarity search
- Connected to **Oracle Autonomous Database** for persistent structured storage with hybrid search capabilities across legal document metadata

CI/CD & Infrastructure-as-Code

- Engineered `build_and_push_ocir.sh` automation script to compile Docker images and push to **Oracle Cloud Infrastructure Registry (OCIR)**
- Created `deploy_backend_oci.sh` deployment runbook for pulling and executing the full stack on OCI compute instances
- Authored `docker-compose.oci.yml` for multi-service orchestration with health checks and restart policies

Securing Cloud-Native Workloads

- Configured **Nginx reverse proxy** (`nginx.conf`) to shield internal backend ports and enforce TLS termination
- Implemented strict **CORS protection** at the reverse proxy layer with origin whitelisting
- Deployed **client-side encryption** (`evidenceCrypto.ts`) for sensitive legal document evidence with E2EE protocols
- Managed database credentials through **environment-isolated secret management** with OCI Vault integration

Cost Optimization and Monitoring

- Leveraged **Oracle Autonomous DB** auto-scaling features to prevent compute overhead during idle periods
- Maintained high-speed retrieval performance while minimizing resource allocation during low-traffic windows
- Configured automated backup policies and failover mechanisms ensuring consistent availability

2. Interview.io AI Prep Platform: Multi-Cloud & Cost Optimization

Architecting Scalable Infrastructure

- Built cloud-native architecture leveraging **Vercel** edge network for global frontend distribution
- Deployed **Node.js/Express.js** backend with connection pooling for concurrent user handling
- Integrated **MongoDB Atlas** managed cloud database for session persistence and user data
- Established **WebSocket** channels for real-time streaming between frontend and backend services

CI/CD & Infrastructure-as-Code

- Implemented **Hybrid AI routing architecture** with automated health checks for primary/fallback model selection
- Configured continuous deployment pipeline through Vercel's Git integration for zero-downtime releases
- Designed stateless backend services enabling horizontal scaling without session affinity

Securing Cloud-Native Workloads

- Secured user identity and access management using **Firebase Authentication** with secure token validation
- Configured **AWS IAM policies** following least-privilege principles for granular access control
- Hardened **AWS S3 bucket integrations** with presigned URLs and server-side encryption for resume storage
- Implemented CORS and rate limiting at the API gateway layer

Cost Optimization and Monitoring

- Engineered intelligent **Oracle Caching System** for roadmap generation requests, achieving **70%+ cache hit rate**
- Reduced LLM API compute costs by **40-60%** through strategic response caching and TTL management
- Implemented automated failover switching to Cohere models during GPT-4 rate limiting, ensuring **99.9% uptime**

3. AI Receptionist: Serverless & Event-Driven Cloud Workflows

Architecting Scalable Infrastructure

- Built entirely **serverless, event-driven** integration topology eliminating always-on compute costs
- Utilized **n8n** as centralized orchestrator handling asynchronous webhook traffic from **Vapi** voice layer
- Designed stateless workflow functions triggered exclusively by external events (calendar bookings, voice calls, messages)

CI/CD & Infrastructure-as-Code

- Configured webhook-based deployment model where each workflow update triggers automatically through n8n's API
- Implemented version-controlled workflow definitions stored as JSON configurations in Git
- Established monitoring dashboards for workflow execution metrics and error tracking

Securing Cloud-Native Workloads

- Handled secure **OAuth/API key handshakes** across multiple cloud perimeters including Google Workspace APIs
- Configured **Google Cloud service accounts** with domain-wide delegation for calendar and Gmail access
- Implemented secure credential rotation for **WhatsApp Business API** tokens and DeepSeek API keys
- Enforced environment variable isolation between production and staging webhook endpoints

Cost Optimization and Monitoring

- Adopted pure **serverless/pay-as-you-go** billing model with compute utilized only during webhook execution
- Achieved near-zero idle infrastructure costs by leveraging stateless event-driven workflows
- Configured automated retry policies with exponential backoff to handle transient API failures gracefully

Technical Cloud Toolkit Summary

Category	Tools	Applications
Cloud Providers	Oracle Cloud Infrastructure (OCI), AWS (S3), Vercel, MongoDB Atlas, Google Cloud	Production deployments, storage, serverless functions
Infrastructure & Deployment	Docker, Docker Compose, Bash/Shell Scripting, Nginx, OCIR	Containerization, orchestration, automation, reverse proxy
Security & Identity	Firebase Auth, AWS IAM, Client-side Cryptography, Oracle Vault	Authentication, access control, encryption, secret management
Webhooks & Serverless	n8n, REST APIs, Event-driven architectures	Workflow orchestration, API integrations, async processing

Performance Impact Metrics

Project	Infrastructure Strategy	Key Result	Business Impact
Vidhoor	OCI Multi-container + Nginx	Secure isolated deployment	Protected legal data with E2EE
Interview.io	Multi-cloud + Caching	70%+ cache hit rate	40-60% API cost reduction
Interview.io	Hybrid AI Failover	99.9% uptime	Zero manual intervention required
Receptionist	Serverless n8n	Pay-per-execution	Near-zero idle costs